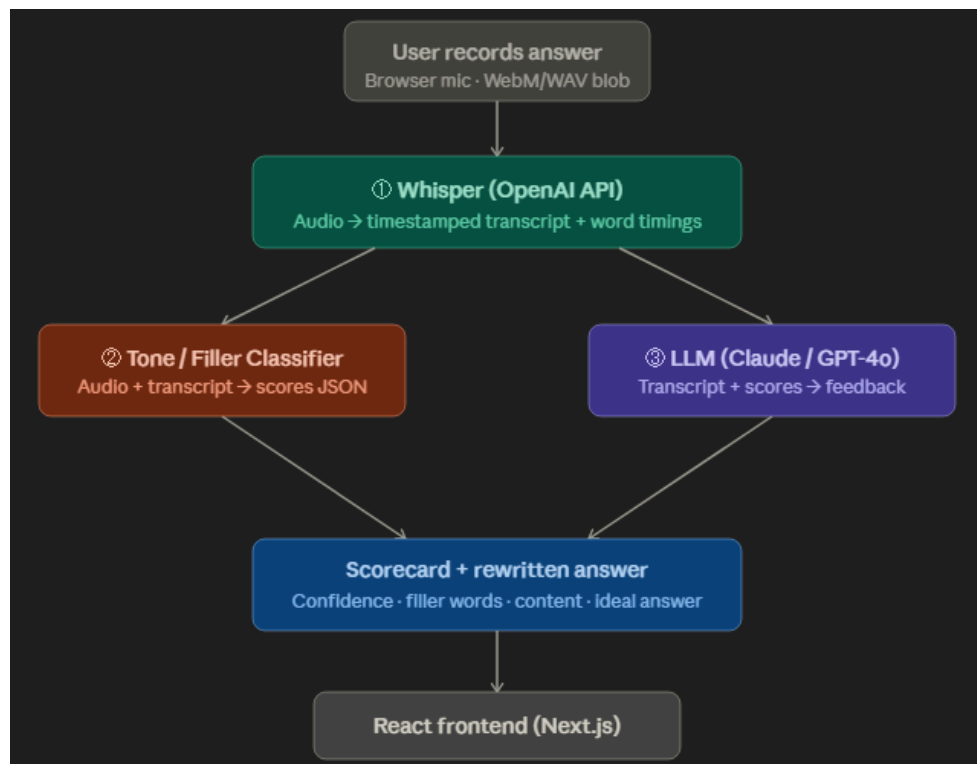


Interview Coach — Project Proposal



The components

① Whisper (OpenAI API) — Speech-to-Text

- **What it is:** OpenAI's [whisper-1](#) model, called via their hosted API
- **Input:** A raw audio recording (WebM or WAV blob) of the user answering an interview question
- **Output:** A timestamped transcript with word-level timings, used downstream by both the classifier and the LLM
- **Where it runs:** Hosted API (OpenAI)
- **Why this one:** Best accuracy on spontaneous speech with filler words, accents, and hesitations. Simpler to integrate than Deepgram and more accurate on natural speech than Web Speech API

② Audeering wav2vec2 — Tone & Delivery Analyzer

- **What it is:** [audeering/wav2vec2-large-robust-12-ft-emotion-msp-dim](#), a pretrained model hosted on HuggingFace that outputs continuous arousal, valence, and dominance scores from raw audio. Dominance maps directly to perceived confidence. Filler word counts are extracted separately from the Whisper transcript via keyword matching ("um", "uh", "like", "you know", "sort of")

- **Input:** The raw audio clip + the Whisper transcript
- **Output:** A scores JSON: `{ confidence: 0.72, arousal: 0.55, filler_words: ["um", "uh", "like"], filler_count: 6, pace_wpm: 142 }`
- **Where it runs:** Local inference on the FastAPI backend via HuggingFace `transformers` (free, no API key needed)
- **Why this one:** Pretrained on a large, diverse speech dataset, no training required, runs locally for free, and the dominance dimension is a meaningful proxy for interview confidence specifically

③ LLM (Claude claude-sonnet-4-20250514 or GPT-4o) — Feedback Generator

- **What it is:** A large language model called via API with a structured prompt
- **Input:** The full transcript + the classifier's scores JSON + the original question text
- **Output:** Structured feedback: what the candidate said well, what to cut, a rewritten model answer, and a 1–10 content score
- **Where it runs:** Hosted API (Anthropic or OpenAI)
- **Why this one:** Handles open-ended qualitative judgment that a classifier can't — "did this answer actually address the question?" — while the classifier handles quantitative delivery metrics

Architecture

The data flow is the same as before:

1. User selects a question and records their answer in the browser
2. Audio blob → Whisper API → timestamped transcript
3. Audio + transcript → Audeering model → delivery scores JSON
4. Transcript + scores + question → LLM → feedback object
5. Frontend renders the scorecard

Stack

- **Frontend:** Next.js (React) — handles recording via `MediaRecorder` API, renders the scorecard UI
- **Backend:** FastAPI (Python) — orchestrates the three components, runs the Audeering model locally
- **Model:** HuggingFace `transformers` + `audeering/wav2vec2-large-robust-12-ft-emotion-msp-dim`
- **APIs:** OpenAI (Whisper), Anthropic or OpenAI (LLM)
- **Deployment target:** Vercel (frontend) + Render or Fly.io (backend)

Concerns and risks

1. **Latency** — Three sequential calls with a local model inference in the middle could push feedback time past 10 seconds. May need to run the Audeering model and Whisper in parallel where possible, and stream the LLM response
2. **Backend memory** — wav2vec2-large is ~1.2GB. Free tiers on Render/Fly may not have enough RAM. Worth checking deployment constraints early — may need to use the [wav2vec2-base](#) variant instead
3. **MediaRecorder compatibility** — Browser audio formats vary by OS/browser. Need to confirm consistent capture or add server-side transcoding (ffmpeg) before passing to Whisper
4. **Dominance as a confidence proxy** — The Audeering model wasn't trained specifically on interview speech. The dominance score is a reasonable proxy but may not always correlate with what a human interviewer would call "confident." Worth sanity-checking on a few recordings early
5. **Cost** — Whisper and the LLM API are the only paid components. Should be negligible for demo usage, but worth setting a budget cap

Who's doing what (*fill in names*)

Area	Owner
Frontend (recording UI, scorecard display)	Niko
Backend API + orchestration	All
Whisper integration	Daniel
Audeering model integration + scoring logic	Mathew
LLM prompt engineering + output parsing	Jacob